

Artificial Intelligence Concepts and Machine Learning Techniques (AI – 103)

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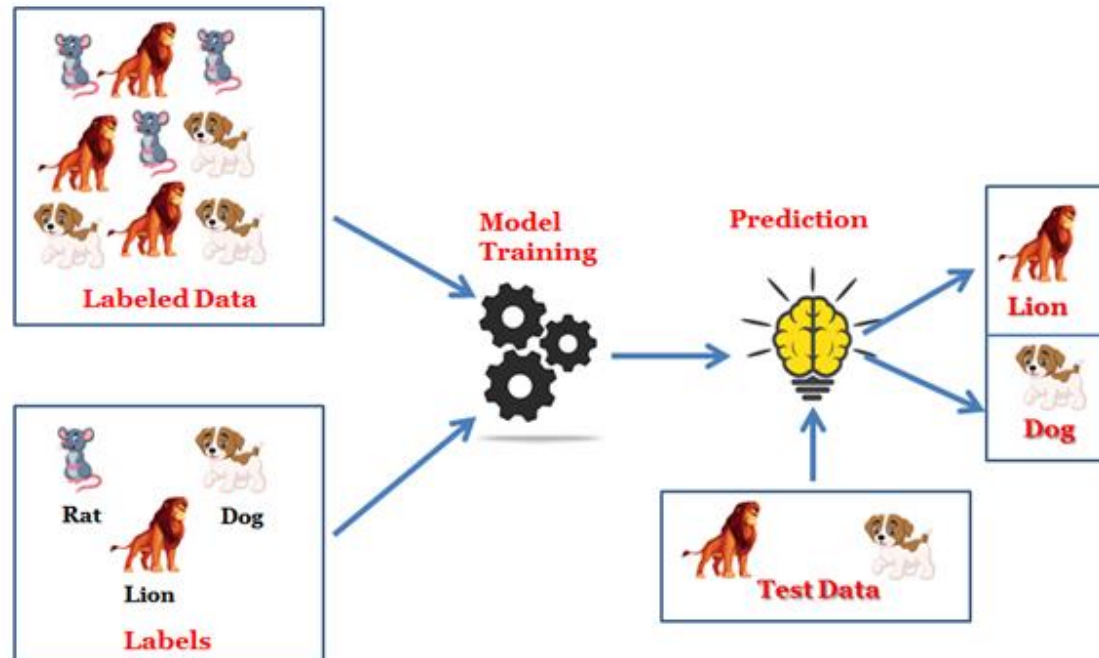
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Supervised Machine Learning Models

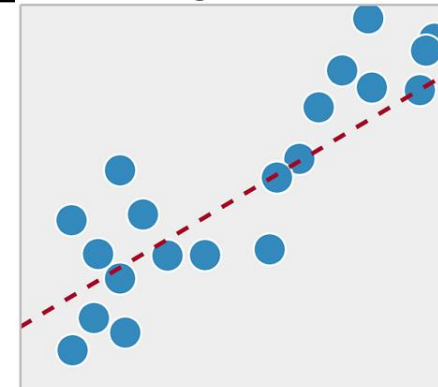
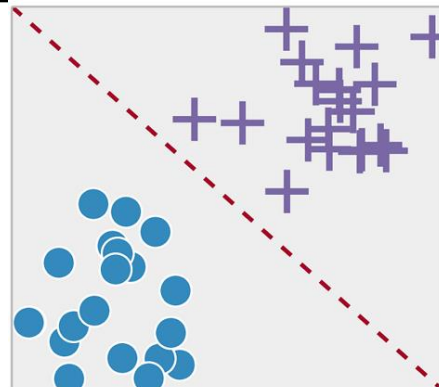
Three minicourses
- supervised learning



Supervised ML



Classification Algorithm	Regression Algorithm
The output variable is discrete in the Classification SML problem.	The output variable has to be a real value or continuous in nature.
Classification algorithm helps in mapping the input value with the output variable which is discrete in nature.	Regression algorithm helps to map the input value and the continuous output variable.
Classification algorithms are only used for data that is discrete.	Regression algorithms are only used for data that is continuous.
Finds the decision boundary which helps in classifying the dataset into different classes in Classification.	Finds the best fit for accurately predicting the output data in Regression.
Classification algorithms are useful for problems like identifying spam emails, recognition of speech, identifying cancer cells, etc.	Regression algorithms help in solving the problems like predicting the weather or the price of Houses in a neighbourhood according to a real estate trend.



Supervised learning overview

- **Supervised learning model** = mapping from one or more inputs to one or more outputs
- Model is a mathematical equation
- Computing the outputs from the inputs = **inference**
- Example:
 - Input is age and milage of secondhand Toyota
 - Output is estimated price of car

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- Parameters affect outcome of equation

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Supervised learning overview

- **Supervised learning model** = mapping from one or more inputs to one or more outputs
- Model is a family of equations
- Computing the outputs from the inputs = **inference**
- Model also includes **parameters**
- Parameters affect outcome of equation
- **Training** a model = finding parameters that predict outputs “well” from inputs for a **training dataset** of input/output pairs

Notation

- For simplicity, we assume that both the input $\mathbf{x}$ and output $\mathbf{y}$ are vectors of a predetermined and fixed size.

Input: $\mathbf{x}$

Output: $\mathbf{y}$

- To make the prediction, we need a model $\mathbf{f}[\bullet]$ that takes input $\mathbf{x}$ and returns $\mathbf{y}$

Model: $\mathbf{y} = \mathbf{f}[\mathbf{x}]$

Example

- Input: $\mathbf{x} = \begin{bmatrix} \text{age} \\ \text{mileage} \end{bmatrix}$ ← Structured or tabular data

- Output: $y = [\text{price}]$

- Model: $y = \mathbf{f}[\mathbf{x}]$

Model

- Parameters: ϕ  The choice of parameters determines the particular relation between input and output.
- Model : $y = f[x, \phi]$
- *learning or training* a model  we mean that we attempt to find parameters ϕ that make sensible output predictions from the input.
 - ➔ We learn these parameters using a *training dataset* of / pairs of input and output examples $\{x_i, y_i\}$.
 - ➔ the aim is to select parameters that map each training input to its associated output as closely as possible.
 - ➔ quantify the degree of mismatch in this mapping with the *loss* L .

Loss function

- Training dataset of I pairs of input/output examples:

$$\{\mathbf{x}_i, \mathbf{y}_i\}_{i=1}^I$$

- **Loss function** or **cost function** measures how bad model is:

$$L[\phi, f[\mathbf{x}_i, \phi], \{\mathbf{x}_i, \mathbf{y}_i\}_{i=1}^I]$$

or for short:

$$L[\phi]$$

← Returns a scalar that is smaller when model maps inputs to outputs better

Training

- Loss function:

$$L[\phi]$$

- Find the parameters that minimize the loss:

$$\hat{\phi} = \underset{\phi}{\operatorname{argmin}} [L[\phi]]$$

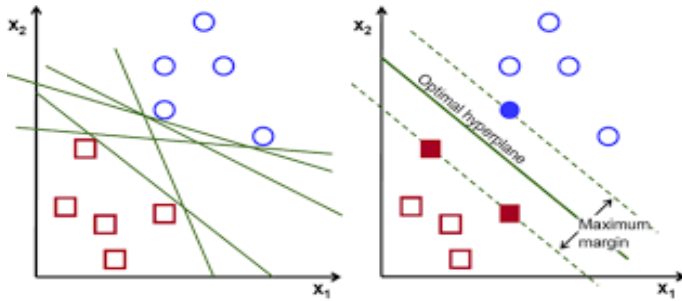
Testing

- To test the model, run on a separate **test dataset** of input / output pairs
- See how well it **generalizes** to new data

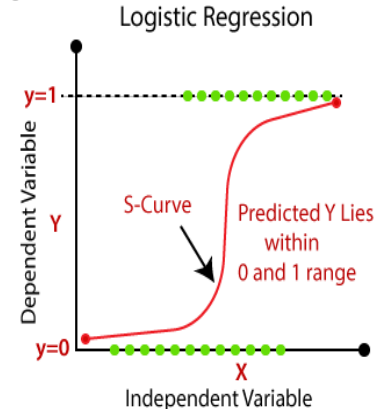
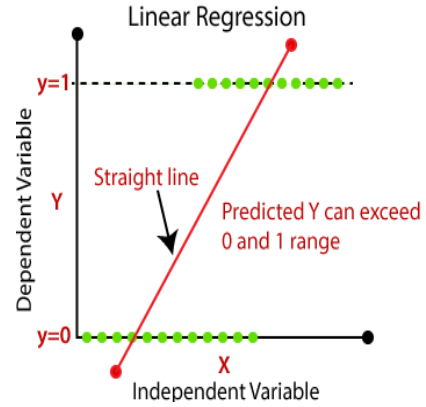
Goal of Supervised Learning?

- Minimize the probability of model prediction errors (Loss Function) on future data
- Two Competing Methodologies
 - Build one really good model
 - Traditional approach
 - Build many models and average the results
 - Ensemble learning (more recent)

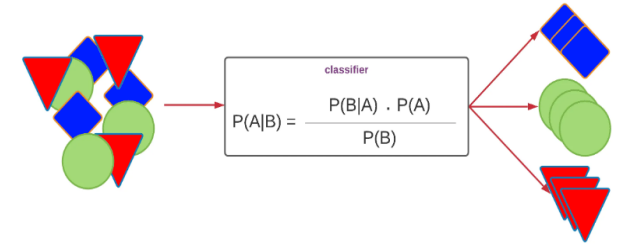
Support Vector Machine



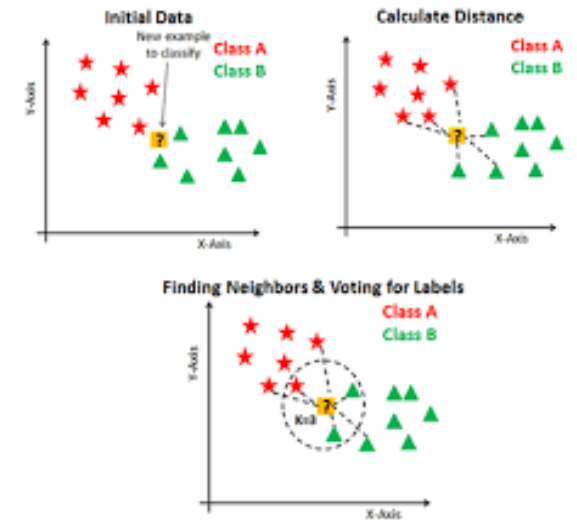
Regression



Naive Bayes



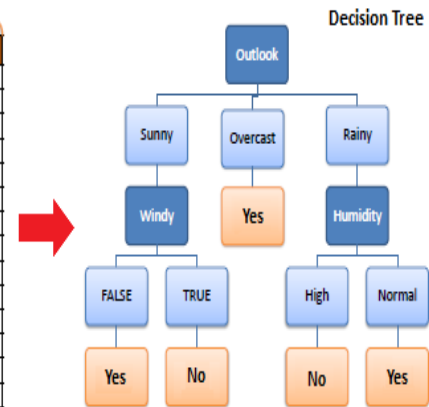
K Nearest Neighbor



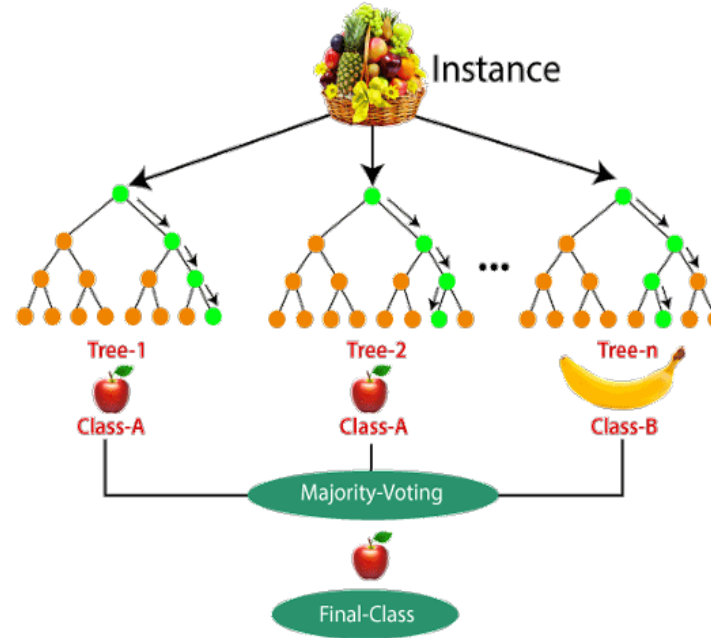
One Model

Decision Tree

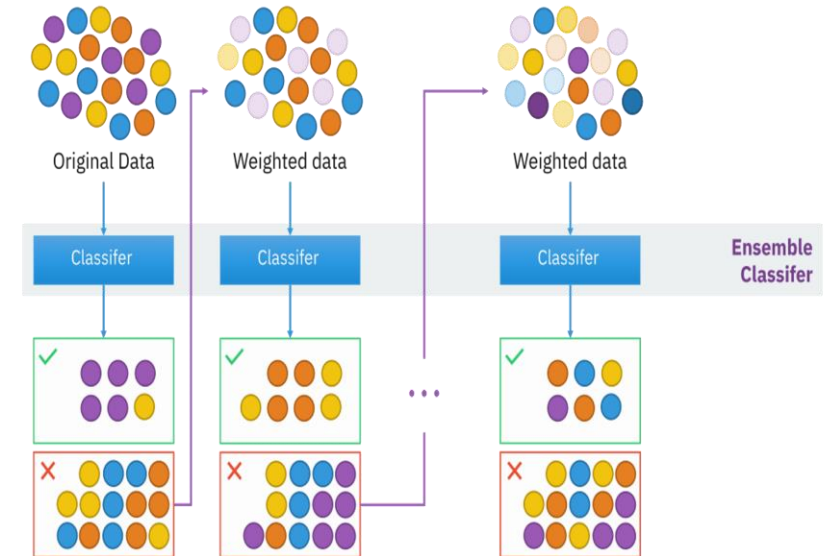
Predictors				Target
Outlook	Temp	Humidity	Windy	Play Golf
Rainy	Hot	High	False	No
Rainy	Hot	High	True	No
Overcast	Hot	High	False	Yes
Sunny	Mild	High	False	Yes
Sunny	Cool	Normal	False	Yes
Sunny	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Rainy	Mild	High	False	No
Rainy	Cool	Normal	False	Yes
Sunny	Mild	Normal	False	Yes
Rainy	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Sunny	Mild	High	True	No



Random Forest



Adaptive Boosting (AdaBoost)



Many models (Ensemble Models)

<https://www.almabetter.com/bytes/tutorials/data-science/adaboost-algorithm>

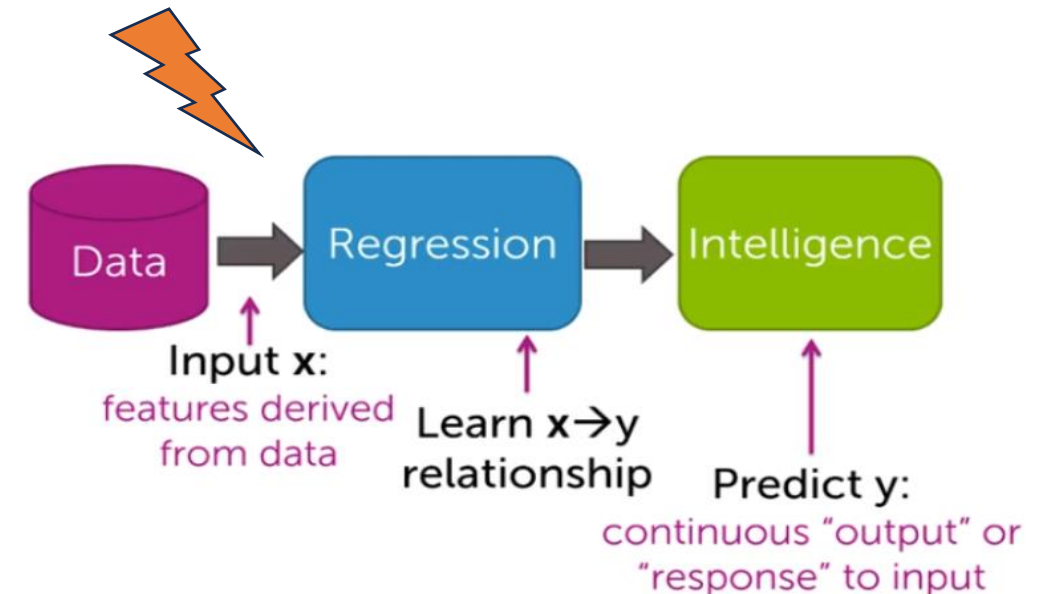
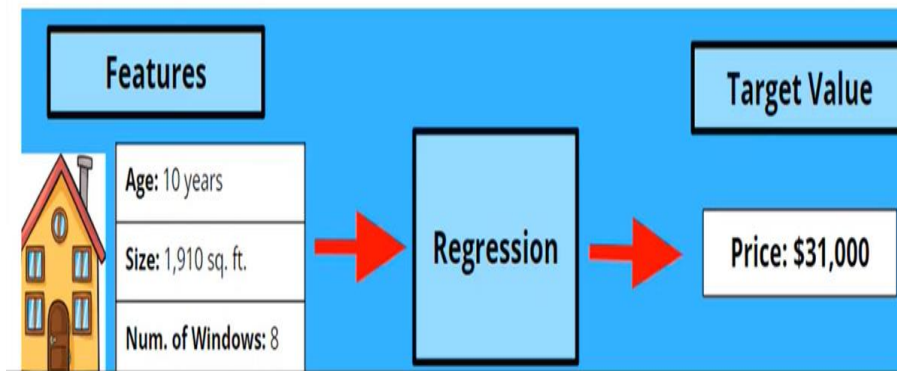


Regression Models



Regression in ML

- **Regression** is a supervised machine learning technique used to estimate the relationship between features and their label (i.e., target). Features can be categorical or numerical and Target variable must be **real number or continuous**.
- **The Regression algorithm's task** is finding the mapping function so we can map the input variable of "x" to the continuous output variable of "y."
- **Ex: predict** the house price, market trends, ...



Domain of applications



Predictive Modeling: predicting an output variables based on one or more input variables. Ex: predict house price based on various features such as square footage, number of bedrooms, and location.



Correlation analysis: understanding the strength and relationship between two or more variables. Ex: understanding the relationship between age and cholesterol levels.



Trend analysis: analyzing trends over time or across different levels of an independent variable or analysis of the future based on past performance results. Ex: A team of financial analysts may review sales patterns to determine whether they are increasing or decreasing.

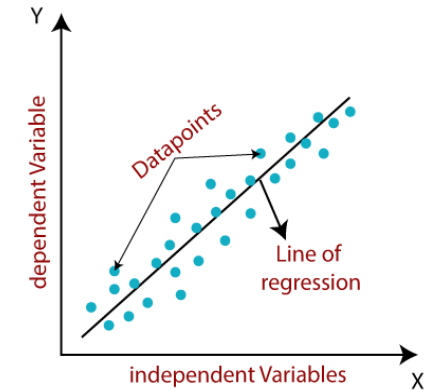


Economic Forecasting: predicting the future conditions of the economy using the combination of important indicators. Ex: stock price changes.

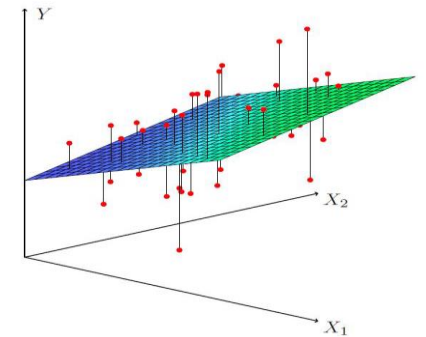
Types of Regression

The most common categories are:

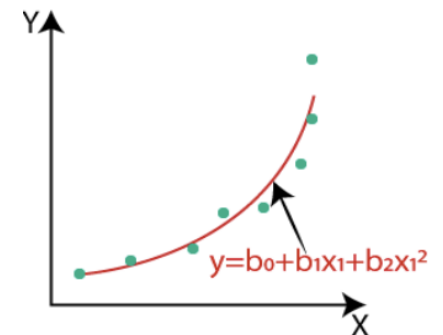
Linear regression: is the basic form, it models the relationship between the dependent and one/more independent variables using linear equations. It assumes a linear relationship between the variables.



Multiple Regression: it extends the linear by involving more than one independent variable to predict the dependent variable. It can be used where multiple factors have influence on the output.



Polynomial Regression: when the relationship between variables are not linear, a polynomial equation (e.g., quadratic or cubic) used to capture more complex patterns.



Linear Regression (LR)

Linear Regression is a statistical tool that gives us the ability to estimate the mathematical relationship between a dependent variable (usually denoted by y) and one or more independent variable/s (usually denoted by x).

- **Dependent variable:** The variable **we wish** to predict or explain. Also, it known as response, outcome.
- **Independent variable:** The variable **used to** predict or explain the dependent variable. Also, it known as predictor or explanatory.

LR can be used in various purposes in different domains such as statistics, social sciences, economics, and data analysis.

Simple Linear Regression

Simple LR concerns with the study of only one predictor variable.

The simple linear regression equation provides an **estimate** of the population regression line

- Only **one independent variable**, X
- Relationship between X and Y is described by a **linear function**.
- Changes in Y are assumed to be related to changes in X.

Estimated (or predicted) Y value for observation i

Estimate of the regression intercept

Estimate of the regression slope

Value of X for observation i

$$\hat{Y}_i = b_0 + b_1 X_i$$

Terms
 $b_0 \approx \phi_0$
 $b_1 \approx \phi_1$

Multiple Linear Regression

Multiple LR concerns with the study of two or more predictor variables.

Multiple Regression Model with k Independent Variables:

Examine the linear relationship between one dependent (Y) & two or more Independent variables (X_i).

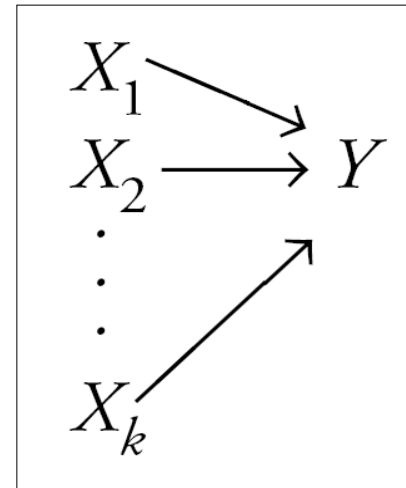
Estimated (or predicted) value of Y

Estimated intercept

Estimated slope coefficients

$$\hat{Y}_i = b_0 + b_1 X_{1i} + b_2 X_{2i} + \dots + b_k X_{ki}$$

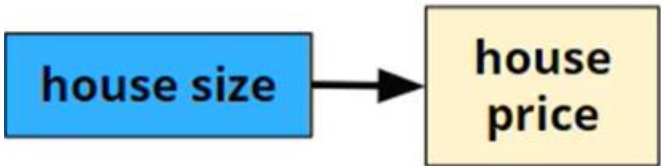
The coefficients of the multiple regression model are estimated using sample data





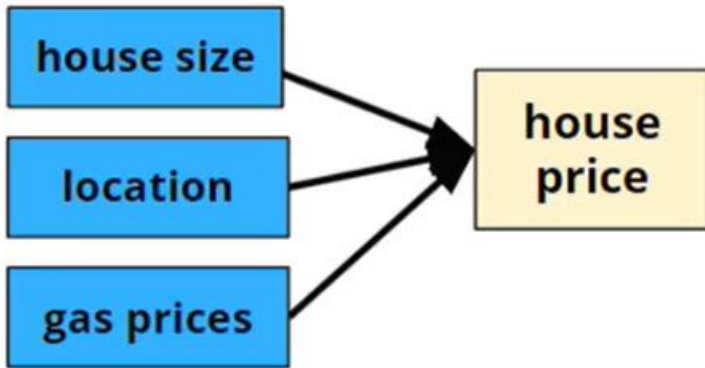
Example

Simple Regression



1 feature,
1 target

Multiple Regression



Multiple features,
1 target

Polynomial Linear Regression

It does not depend on the variables, instead, **it depends on the coefficients, which are arranged in a linear fashion.**

Kth order polynomial model of one variable:

The relationship between the independent variable x and the dependent variable y is modeled as an n th degree polynomial.

Estimated
(or predicted)
value of Y

Estimated
intercept

Estimated slope coefficients

$$\hat{Y}_i = b_0 + b_1 X_{1i} + b_2 X_{1i}^2 + \dots + b_k X_{1i}^k$$

It is a nonlinear relationship between the value of x and the corresponding conditional mean of y , denoted $E(y | x)$

The coefficients of the polynomial regression model are estimated using sample data

The key factors in linear regression

The goal is to find the best line that fits the observed data

Dependent Variable

Population Y intercept

Population Slope Coefficient

Independent Variable

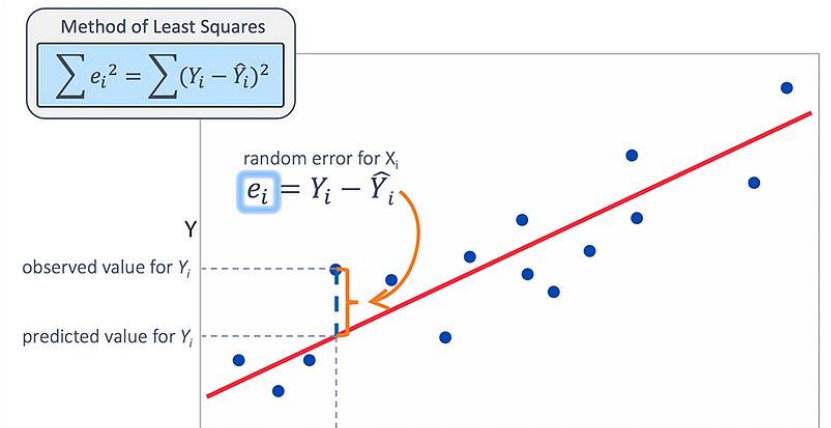
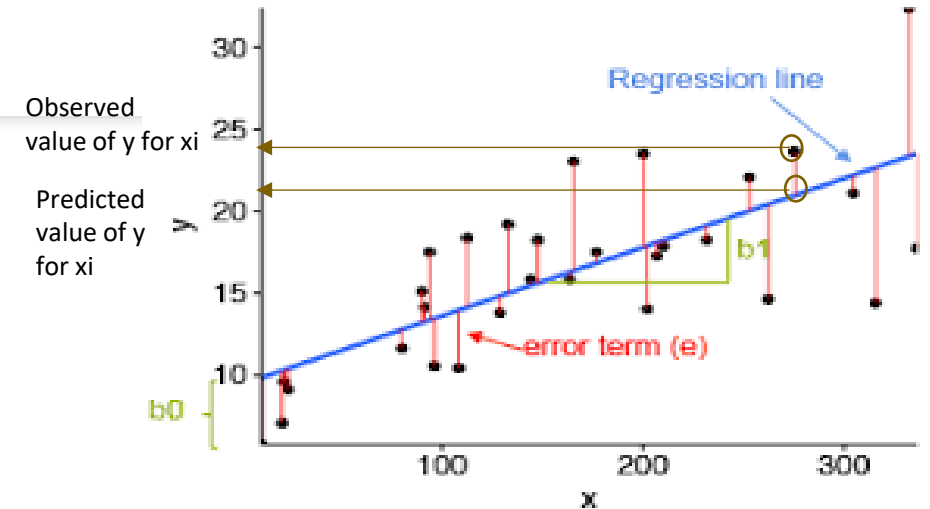
Random Error term

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

Linear component

Random Error component

- The error term also known as residual is defined as the difference between the actual value (observed) and estimated value (predicted).
- b_0 is the estimated mean value of Y when the value of X is zero.
- b_1 is the estimated change in the mean value of Y as a result of a one-unit increase in X.
- Find the **least squared error**.



b_0 and b_1 are obtained by finding the values that minimize the sum of the squared differences between Y and $\hat{Y}$

$$\min \sum (Y_i - \hat{Y}_i)^2 = \min \sum (Y_i - (b_0 + b_1 X_i))^2$$



- Model: 1D linear regression model

$$y = f[x, \phi]$$

↑
dependent variable

$$= \phi_0 + \phi_1 x \leftarrow \text{independent variable}$$

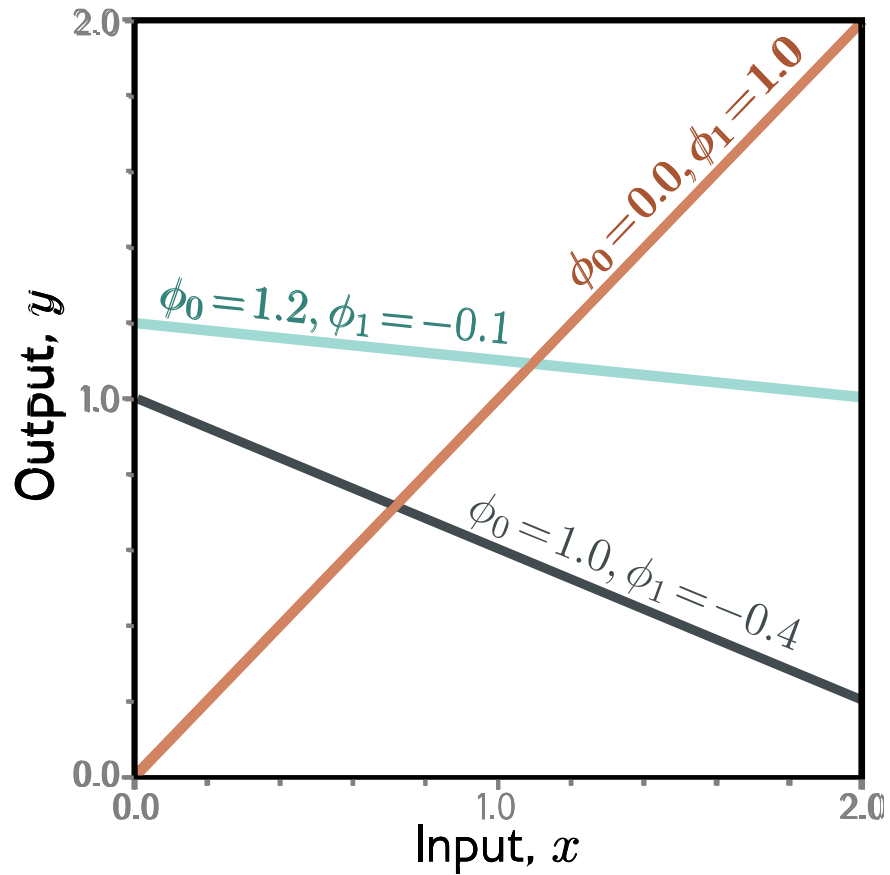
- Parameters

$$\phi = \begin{bmatrix} \phi_0 \\ \phi_1 \end{bmatrix}$$

← y-offset / intercept

← slope

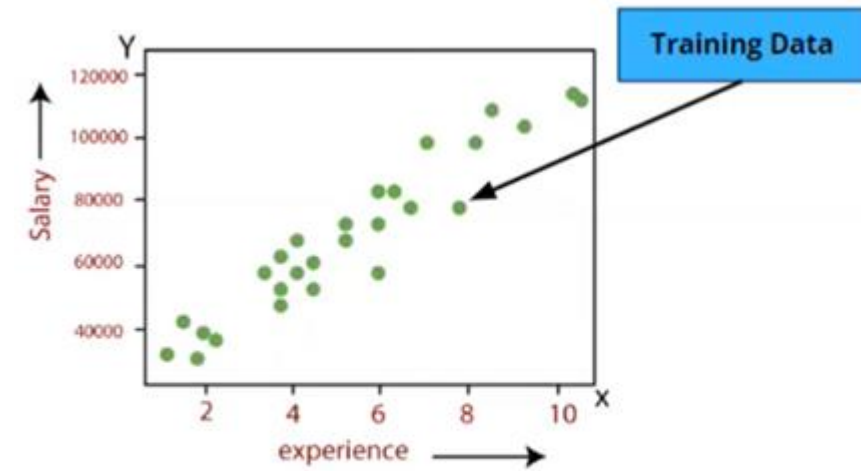
- ϕ_0 is the y-intercept (the estimated mean value of Y when X = 0)
- ϕ_1 is the slope (the estimated change in the mean value of Y as a result of one-unit change in X)



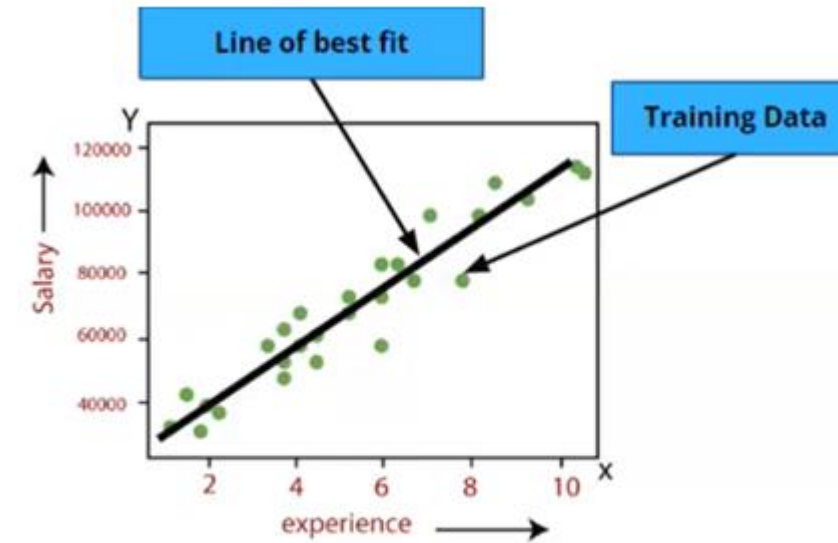
Different choices for the y-intercept ϕ_0 and the slope ϕ_1 change these predictions (cyan, orange, and gray lines).

Example

There is no straight line that can pass all the points

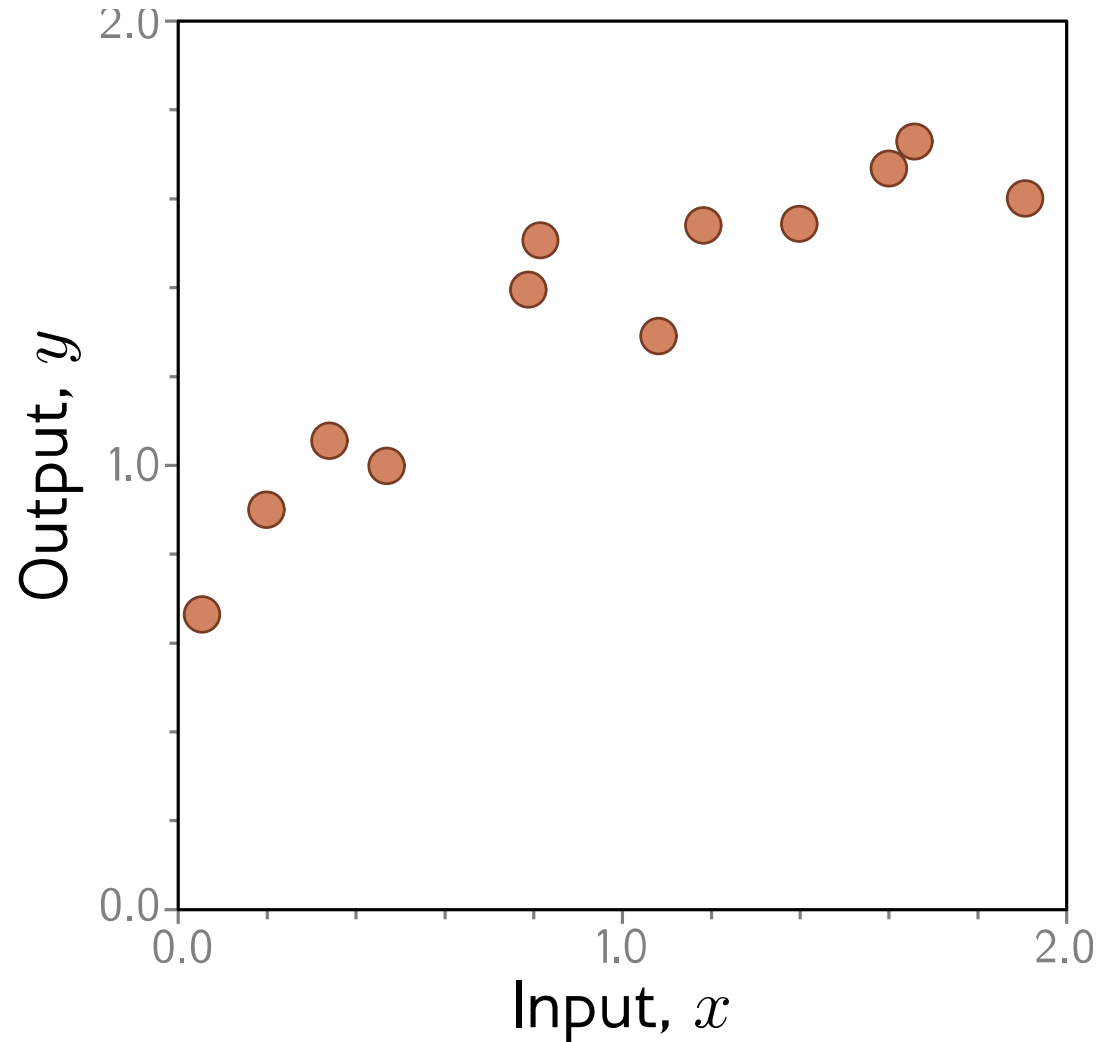


Before Linear Regression



After Linear Regression

Example: 1D Linear regression training data



- We assign a numerical value to each choice of parameters that quantifies the degree of mismatch between the model and the data.
- We term this value the *loss*; a lower loss means a better fit.

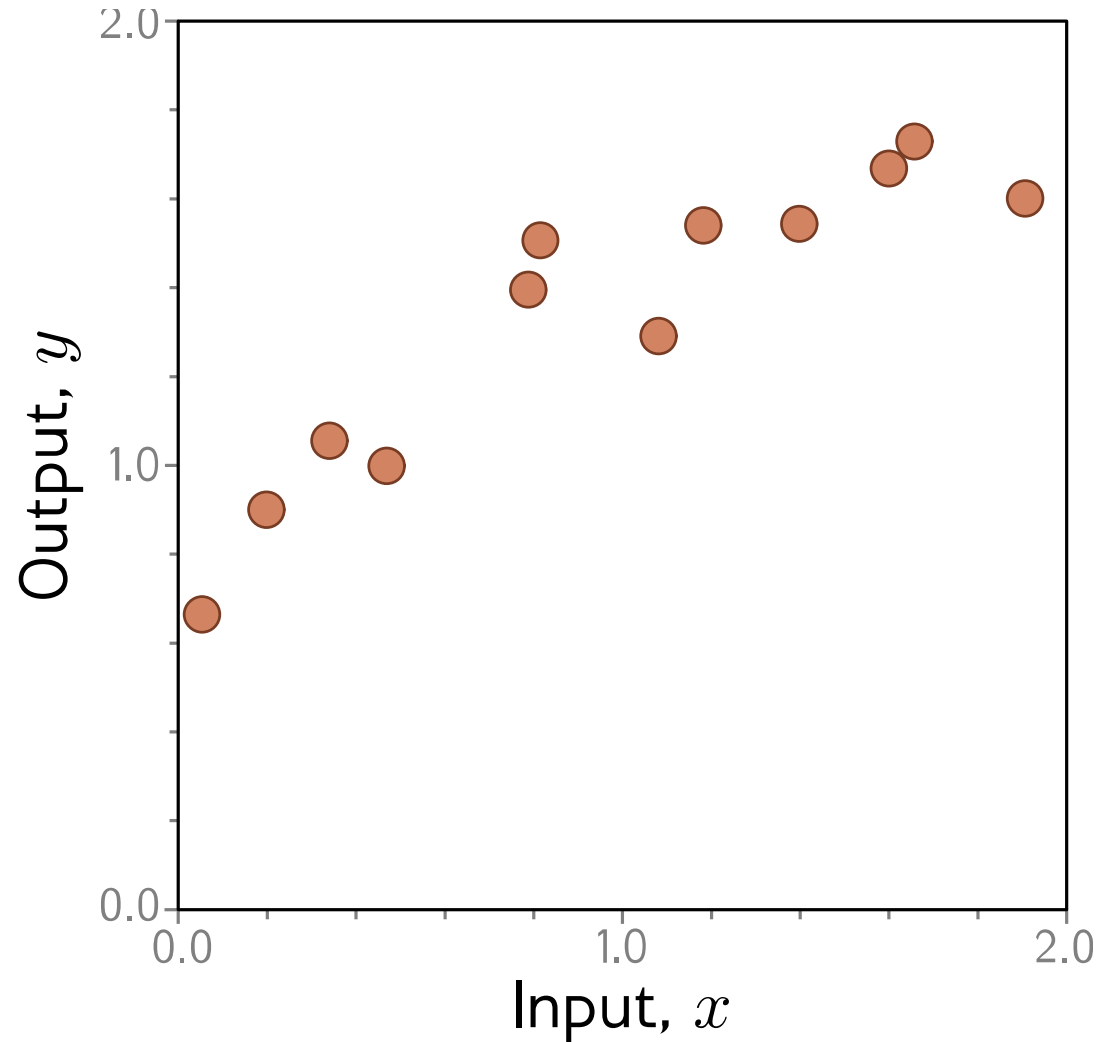


The mismatch is captured by the deviation between the model predictions $f[x_i, \boldsymbol{\varphi}]$ (height of the line at x_i) and the ground truth outputs y_i .



The total mismatch, **training error, or loss** as the sum of the squares of these deviations for all I training pairs.

Example: 1D Linear regression training data

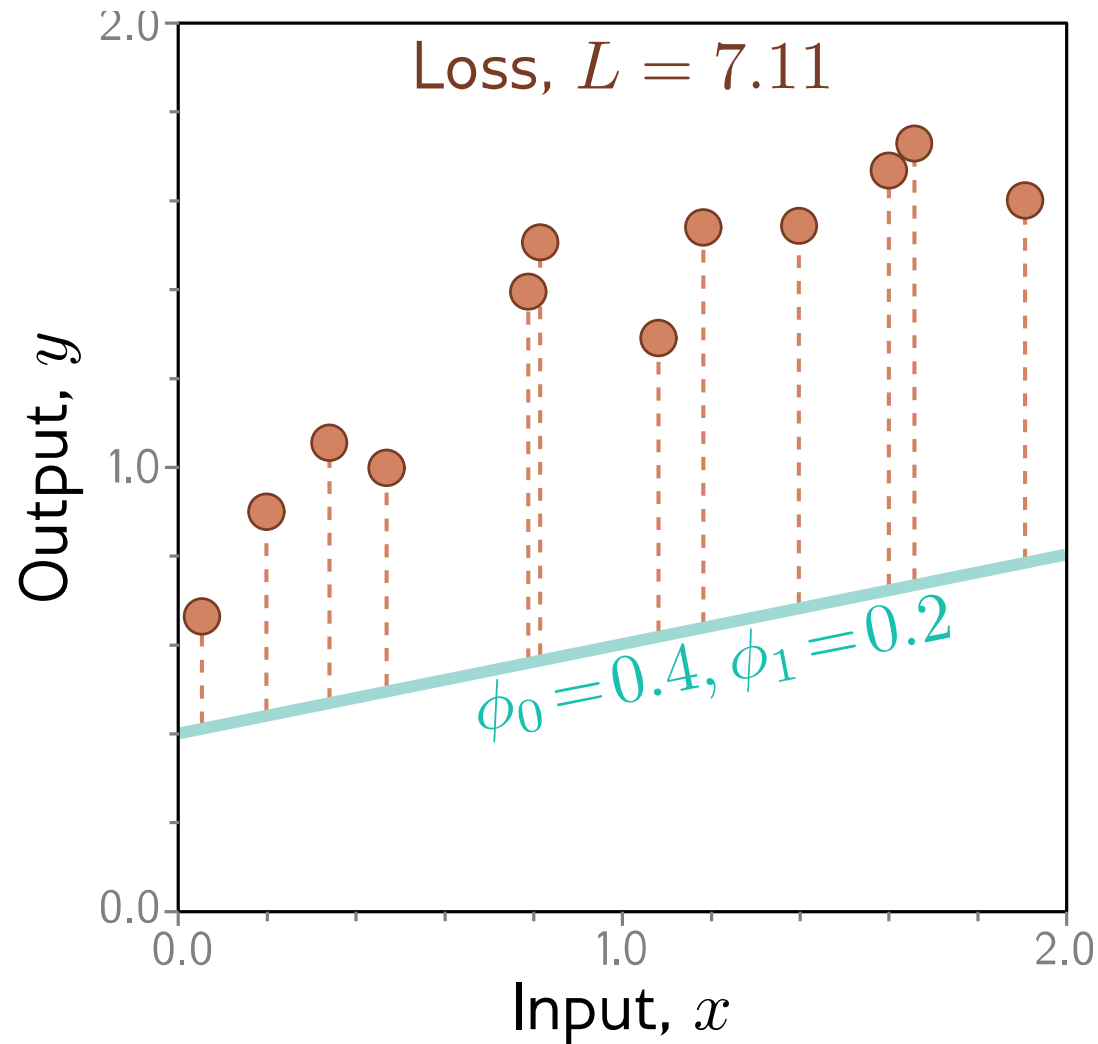


Loss function:

$$\begin{aligned} L[\phi] &= \sum_{i=1}^I (f[x_i, \phi] - y_i)^2 \\ &= \sum_{i=1}^I (\phi_0 + \phi_1 x_i - y_i)^2 \end{aligned}$$

“Least squares loss function”

Example: 1D Linear regression loss function



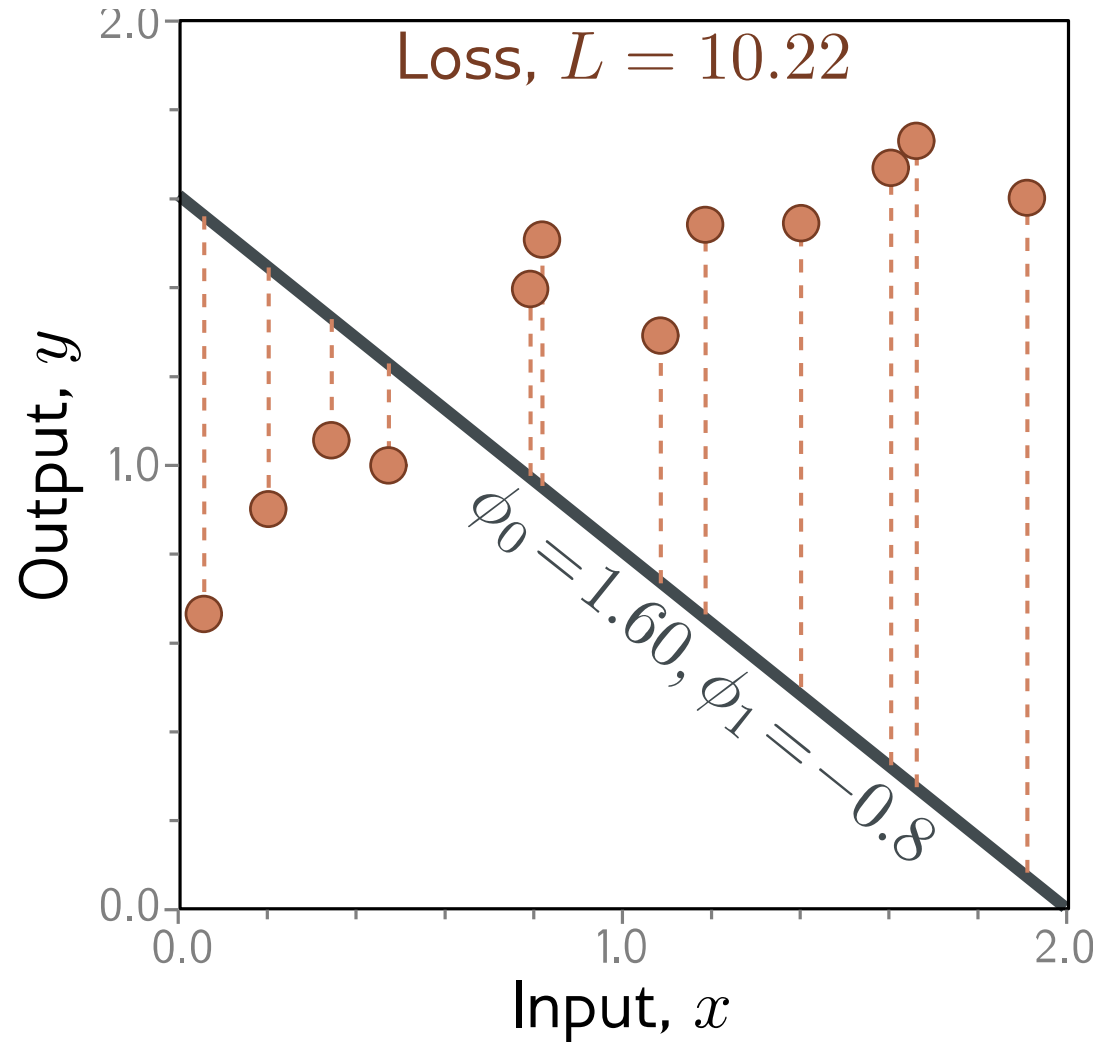
Loss function:

The direction of the deviation

$$L[\phi] = \sum_{i=1}^I (f[x_i, \phi] - y_i)^2$$
$$= \sum_{i=1}^I (\phi_0 + \phi_1 x_i - y_i)^2$$

“Least squares loss function”

Example: 1D Linear regression loss function

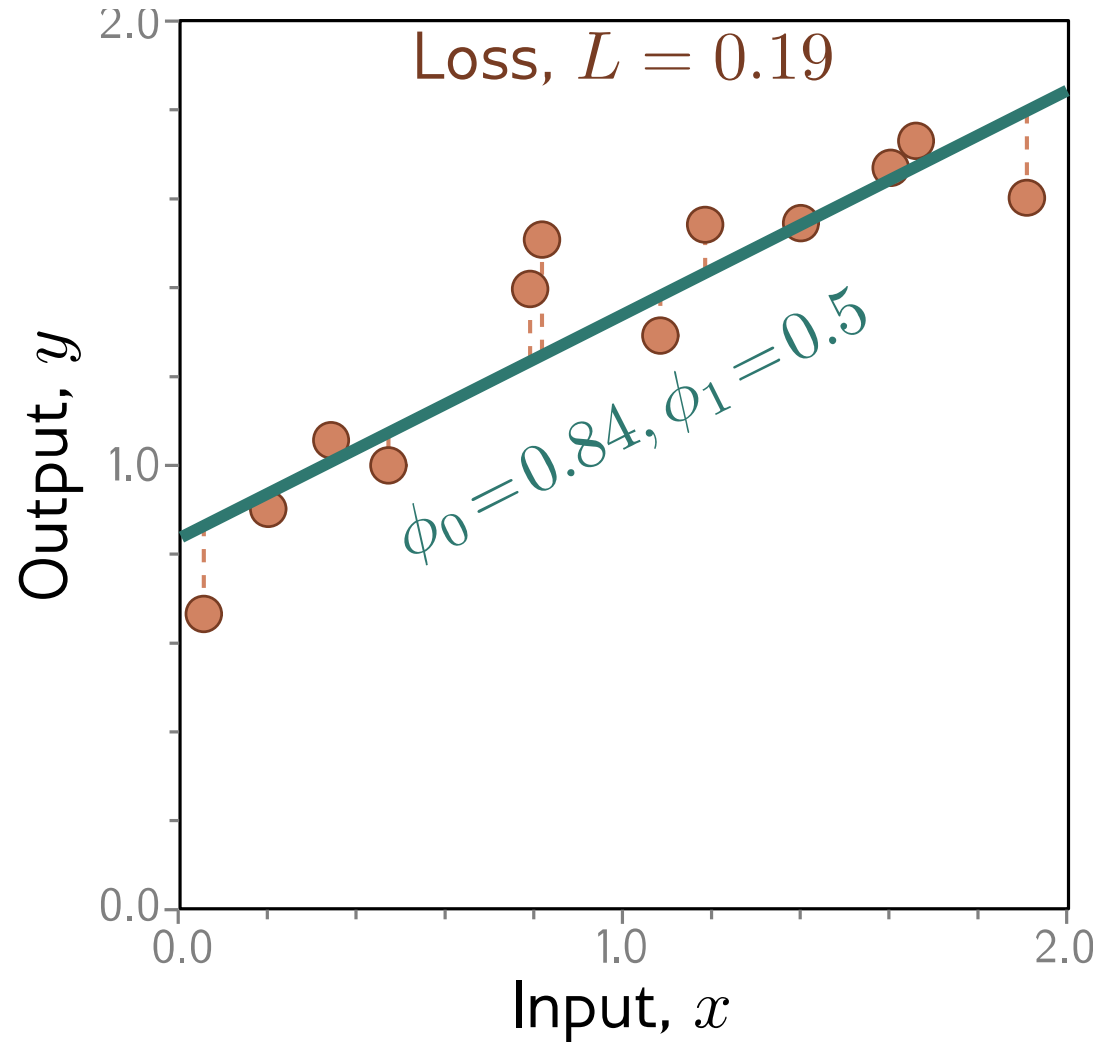


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Example: 1D Linear regression loss function



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“Least squares loss function”

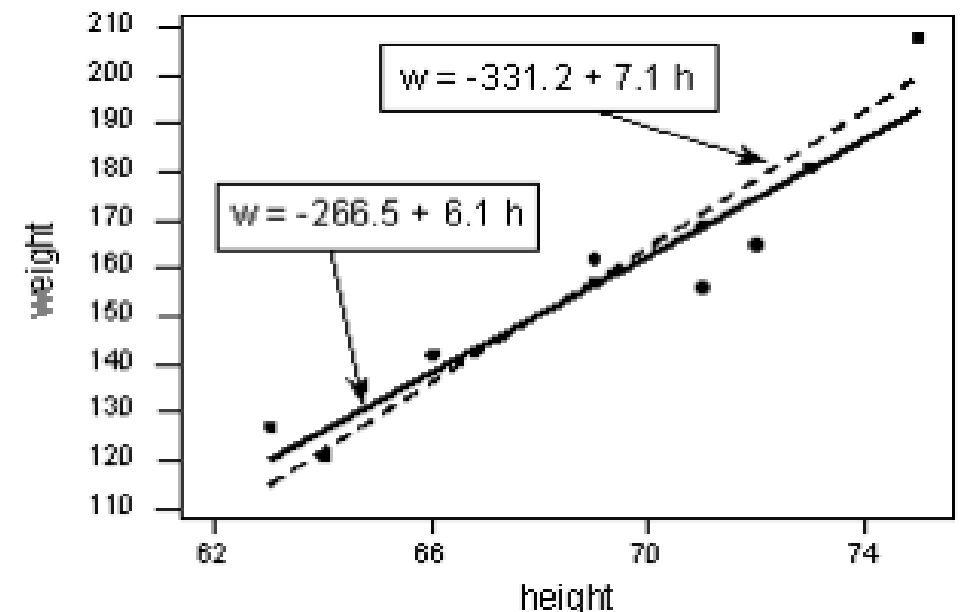
The linear line

The linear line formed as a result of simple regression is $f(x) = b_0 + b_1 x$, and the model formed is $y_i = b_0 + b_1 x_i + \varepsilon_i$. b_0 and b_1 are intercept and slope, which are regression coefficients.

How to find a model that is suitable for the data, or what is the criterion for model suitability?

- Let's examine this concept on an example created from students' height and weight information.
- The 2 lines were fitted to the data of 10 students, but **which one is more suitable?**

An infinite number of lines can fit this data (e.g., solid and dashed) But only one line that provides the **best fit**.



Which is the "Best fitting line"?

Consider the students' data **h** for height and **w** for weight. We should decide which one of them is fittest line.

$$\text{weight} = -266.5 + 6.1 * \text{height}$$

or

$$\text{weight} = -331.2 + 7.1 * \text{height}$$

- y_i is the observed response of experimental unit i
- x_i is the predictor value of experimental unit i
- $\hat{y}_i$ is the predicted/estimated response (fitted value) of experimental unit i

The equation for the best fitting line is: $\hat{y}_i = b_0 + b_1 x_i$

Calculation of "prediction error"

$\hat{y}_i$ can be used to predict the actual value y_i with a prediction error e_i (or residual error) according to the following equation.

$$e_i = y_i - \hat{y}_i$$

- The equation for the best fitting line is $\hat{y}_i = b_0 + b_1x_i$
- We should find the values of b_0 and b_1 that minimize the sum of squared prediction errors.
- Therefore, the goal is to find the value of b_0 and b_1 that minimize the following:

Residual Sum of Squares

$$RSS = Q = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Mathematically, we can label the error sum of squares as the sum of squared distances between the observed data and their mean values based on the model.

$$\hat{y}_i \cong w_i \quad x_i \cong h_i$$

$w = -331.2 + 7.1 h$ (the dashed line)					
i	x_i	y_i	$\hat{y}_i$	$(y_i - \hat{y}_i)$	$(y_i - \hat{y}_i)^2$
1	63	127	116.1	10.9	118.81
2	64	121	123.2	-2.2	4.84
3	66	142	137.4	4.6	21.16
4	69	157	158.7	-1.7	2.89
5	69	162	158.7	3.3	10.89
6	71	156	172.9	-16.9	285.61
7	71	169	172.9	-3.9	15.21
8	72	165	180.0	-15.0	225.00
9	73	181	187.1	-6.1	37.21
10	75	208	201.3	6.7	44.89

Calculate the sum of squared error?

766.5

$$w_1 = -331.2 + 7.1 * 63 = 116.1$$

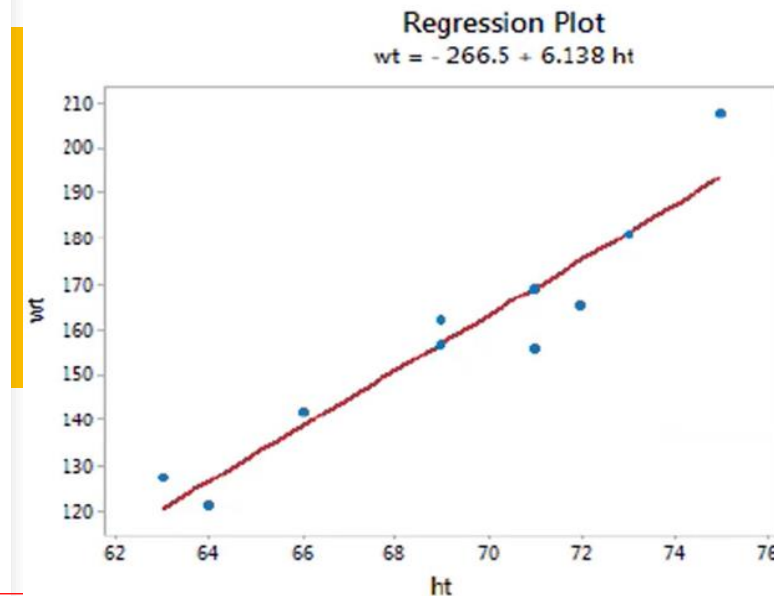
$w = -266.53 + 6.1376 h$ (the solid line)					
i	x_i	y_i	$\hat{y}_i$	$(y_i - \hat{y}_i)$	$(y_i - \hat{y}_i)^2$
1	63	127	120.139	6.8612	47.076
2	64	121	126.276	-5.2764	27.840
3	66	142	138.552	3.4484	11.891
4	69	157	156.964	0.0356	0.001
5	69	162	156.964	5.0356	25.357
6	71	156	169.240	-13.2396	175.287
7	71	169	169.240	-0.2396	0.057
8	72	165	175.377	-10.3772	107.686
9	73	181	181.515	-0.5148	0.265
10	75	208	193.790	14.2100	201.924

597.4

$$w_1 = -266.53 + 6.1376 * 63 = 120.139$$

Which model is the “best”?

The line with the minimum RSS is the best line.



This model is the better model.

Comments

- If we consider the previous approach for finding the equation of the line that minimizes the sum of squared prediction errors, this leads to a tedious task!
- Also, we should have to implement the previous procedure for an infinite number of possible lines, this leads to an impossible task!

In the simple linear regression function: $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$ for $i = 1, \dots, n$

If we actually let $i = 1, \dots, n$, we see that we obtain n equations:

$$\begin{aligned}y_1 &= \beta_0 + \beta_1 x_1 + \epsilon_1 \\y_2 &= \beta_0 + \beta_1 x_2 + \epsilon_2 \\&\vdots \\y_n &= \beta_0 + \beta_1 x_n + \epsilon_n\end{aligned}$$

Well, that's a pretty inefficient way of writing it all out! we can instead formulate the above simple linear regression function in matrix notation:

The goal is minimizing the error vector.

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} + \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{bmatrix} \quad \Leftrightarrow \quad \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} + \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{bmatrix}$$

$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$

$\mathbf{Y}$ (nx1) = $\mathbf{X}$ (nx2) $\boldsymbol{\beta}$ (2x1) + $\boldsymbol{\epsilon}$ (nx1)

$\mathbf{Y}$ is the output/predicted vector (having y_1 to y_n)
 $\boldsymbol{\beta}$ is the model's parameter vector
 $\mathbf{X}$ is the feature vector (having x_1 to x_n)

Simple LR Mathematically

Recall to the simple LR model, the dependent variable or response is related to one independent or regressor variable as in the Equation.

$$y = \beta_0 + \beta_1 x + \epsilon$$

ϵ is a random error term

β_0 (intercept) and β_1 (slope) are regression coefficients

To estimate (β_0, β_1) , we find values that minimize squared error

$$\sum_{i=1}^n (y_i - (\beta_0 + \beta_1 x_i))^2$$

Derivation: method of least squares

Simple LR Mathematically (cont.)

- The expected value of y at each level of x is a random variable: $E(y/x) = \beta_0 + \beta_1 x$
- And assume that each observation y can be described by the model : $y = \beta_0 + \beta_1 x + \epsilon$
- For each:
$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i \quad \text{for } i = 1, \dots, n$$
$$\epsilon_i = y_i - \beta_0 - \beta_1 x_i$$
- Then, the sum of squares of the derivations of the observations from the true regression line is:

$$L = \sum_{i=1}^n \epsilon_i^2 = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2$$

The goal is to minimize this function given the optimal values of β_0 and β_1

Simple LR Mathematically (cont.)

- The least squares estimators of β_0 and β_1 , say, $\hat{\beta}_0$ and $\hat{\beta}_1$ must satisfy the following and equals to zero to get the optimal values of the linear model parameter.

$$\left. \frac{\partial L}{\partial \beta_0} \right|_{\hat{\beta}_0, \hat{\beta}_1} = -2 \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$$

$$\left. \frac{\partial L}{\partial \beta_1} \right|_{\hat{\beta}_0, \hat{\beta}_1} = -2 \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) x_i = 0$$

- Therefore, least square normal equations are :

$$\begin{aligned} n\hat{\beta}_0 + \hat{\beta}_1 \sum_{i=1}^n x_i &= \sum_{i=1}^n y_i \\ \hat{\beta}_0 \sum_{i=1}^n x_i + \hat{\beta}_1 \sum_{i=1}^n x_i^2 &= \sum_{i=1}^n y_i x_i \end{aligned}$$

Simple LR Mathematically (cont.)

- Converting the previous two equations to a matrix notation form , we get the following:

$$\begin{bmatrix} n & \sum_{i=1}^n x_i \\ \sum_{i=1}^n x_i & \sum_{i=1}^n x_i^2 \end{bmatrix} \begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{bmatrix} = \begin{bmatrix} \sum_{i=1}^n y_i \\ \sum_{i=1}^n y_i x_i \end{bmatrix}$$

$X'X$ $X'Y$

Get the inverse $(X'X)^{-1}$  Then least squares estimates $b = (X'X)^{-1}X'Y$

- This matrix comes in the form $A\theta = b$ where the parameter vector $\theta = [\beta_0 \quad \beta_1]^T$
- Therefore, θ can be found by solving the equation $\theta = A^{-1}b$

Consider the height of suds ($y = \text{suds}$) in a standard dishpan was recorded for various amounts of soap ($x = \text{soap}$, in grams)

x_i soap	y_i suds	$x_i \times y_i$ so*su	x_i^2 soap ²
4.0	33	132.0	16.00
4.5	42	189.0	20.25
5.0	45	225.0	25.00
5.5	51	280.5	30.25
6.0	53	318.0	36.00
6.5	61	396.5	42.25
7.0	62	434.0	49.00
---	---	-----	-----
38.5	347	1975.0	218.75

$$b = \begin{bmatrix} b_0 \\ b_1 \end{bmatrix} = (X'X)^{-1} X'Y = ?$$

Example



Calculate the estimated slope and intercept

$$X'X = \begin{bmatrix} 7 & 38.5 \\ 38.5 & 218.75 \end{bmatrix} \quad X'Y = \begin{bmatrix} \sum_{i=1}^n y_i \\ \sum_{i=1}^n x_i y_i \end{bmatrix} = \begin{bmatrix} 347 \\ 1975 \end{bmatrix}$$

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

The inverse of a matrix is found using the following formula:

$$(X'X)^{-1} = \begin{bmatrix} 4.4643 & -0.78571 \\ -0.78571 & 0.14286 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1}$$

$$A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$b = (X'X)^{-1} X'Y = \begin{bmatrix} 4.4643 & -0.78571 \\ -0.78571 & 0.14286 \end{bmatrix} \begin{bmatrix} 347 \\ 1975 \end{bmatrix} = \begin{bmatrix} -2.67 \\ 9.51 \end{bmatrix}$$

That is, the estimated intercept is $b_0 = -2.67$ and the estimated slope is $b_1 = 9.51$

Line of Better Fit: **suds = -2.68 + 9.51 * soap**

Alternative notation

- Get the average values of x and y based on the following Equations:

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

$$\bar{y} = \frac{\sum_{i=1}^n y_i}{n}$$

- Therefore, $\hat{\beta}_0$ and $\hat{\beta}_1$ can be computed according to the following simplified Equations:

estimator for intercept estimator for slope

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

sample mean of observed values for independent variable sample mean of observed values for independent variable

estimate of β_1 (slope) error for single value of y error for single value of x

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

sum of terms for all observed values standardized error for single value of x

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$x = \{3, 4, 5, 6, 7, 9, 13, 16\} \quad y = \{1, 3, 4, 4, 6, 8, 9, 12\}$$

estimate for first term

$$\hat{\beta}_1 = \frac{(1-5.88)(3-7.88)}{(3-7.88)^2} = 1.0$$

estimate for all terms

$$\hat{\beta}_1 = \frac{111.875}{144.875} = \sim 0.772$$

Example

Calculate the estimated slope and intercept

sample mean of y values estimated slope sample mean of x values

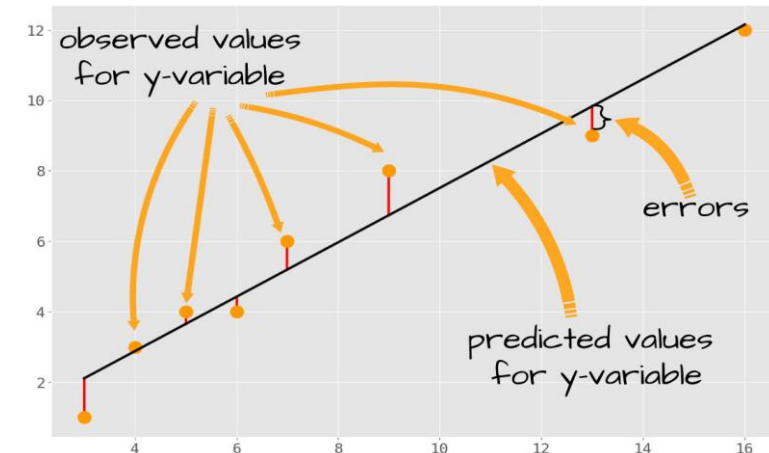
$$\hat{\beta}_0 = \bar{y} - \beta_1 \bar{x}$$

$$\hat{\beta}_0 = 5.88 - .772 * 7.88$$

$$\hat{\beta}_0 = \sim -0.21$$

estimated y-intercept value

Line of Better Fit: $y = -0.21 + 0.772 * x$



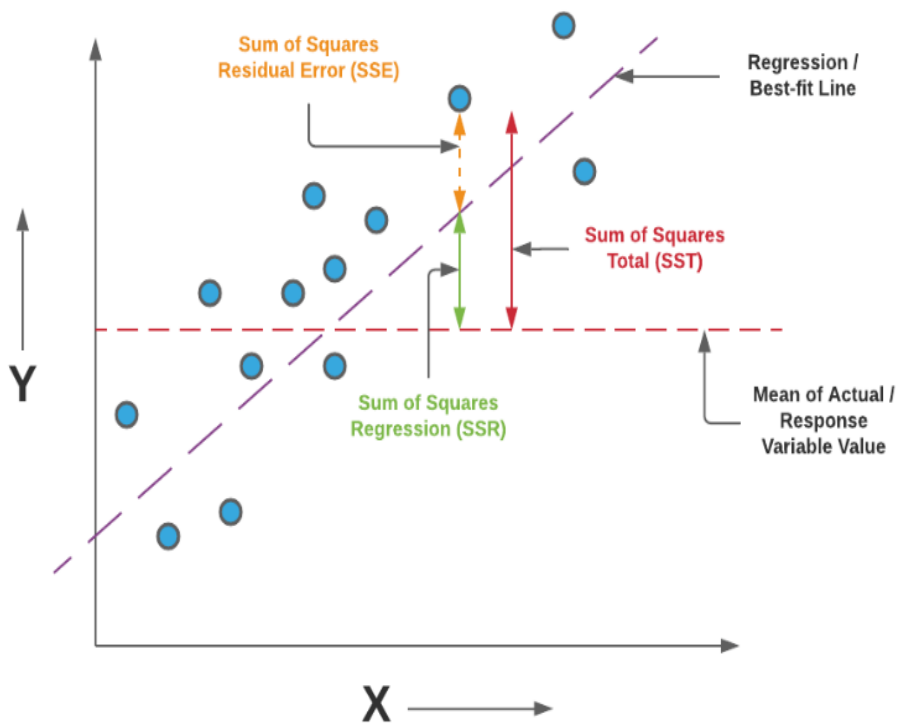
Training

- Finding parameters that minimize loss by randomly selecting initial parameters and improving them by "walking down" the loss function until the gradient is flat.

Testing

- Test with different set of paired input/output data
 - Measure performance / Model Assessment
 - Degree to which this is same as training = **generalization**
- Might not generalize well because
 - Model too simple → **underfitting**
 - Model too complex → **overfitting**

Model Assessment (measures of model fit)



Sum of Square Total (SST): is equal to the sum of the squared difference between actual values related to the response variable and the mean of actual values. “measure how far the data are from the mean”

$$SST = \sum_{i=1}^n (y_i - \bar{y})^2$$

Sum of Square Error (SSE): is the sum of the squared difference between the actual value and the predicted value. “measure how far the data are from the model’s predicted values”

$$SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Sum of Square Regression (SSR): is the sum of the squared difference between the predicted value and the mean of actual values.

$$SSR = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2$$

$$SST = SSR + SSE$$

Model Assessment (measures of model fit)

A well-fitting regression model results in predicted values close to the observed data values. Three statistics are used in Least Squares Estimations (LSE) regression to evaluate model fit: **R-squared**, the overall **F-test**, and the **Root Mean Square Error (RMSE)**.

- **R-squared** (coefficient of determination): is a measure of how good is the regression or best-fit line. It is represented as the ratio of Sum of Squares Regression (SSR) and Sum of Squares Total (SST).

$$\text{R-Squared} = SSR / SST = (SST - SSE) / SST = 1 - (SSE / SST)$$

The greater the value of R-Squared, the better the regression line as higher is the variation explained by the regression line.

- **RMSE**: is the square root of the average squared distance (difference between actual and predicted value).

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

It determines whether there are any large errors or distances that could be caused if the model overestimated the prediction (the model predicted values higher than the actual value) or underestimated the predictions (that is, predicted values less than actual prediction).

Model Assessment (measures of model fit)

A well-fitting regression model results in predicted values close to the observed data values. Three statistics are used in Least Squares Estimations (LSE) regression to evaluate model fit: **R-squared**, the overall **F-test**, and the **Root Mean Square Error (RMSE)**.

- **F-test:** For a multiple regression model with intercept, it tests the null hypothesis and alternative hypothesis.

$$H_0: \beta_1 = \beta_2 = \dots = \beta_{p-1} = 0$$

$$H_1: \beta_j \neq 0, \text{ for at least one value of } j$$

Then based on the confidence interval, the f-test can be computed.

Loss functions vs. cost functions??

Both terms are used interchangeably... But

Loss function : associated with every training sample

Cost function : is the average of the loss function values over all the data samples.

In ML, we optimize the cost function rather than the loss function.

Example:

Absolute error called L1 loss

$$Loss_i = |Y_i - f(x_i)|$$

The corresponding cost function is the average of absolute losses over training samples : Mean Absolute Error (MAE)

$$MAE = \frac{1}{n} \sum_i^n |Y_i - f(X_i)|$$



Test Your Knowledge

MCQ

In linear regression, which of the following represents the *dependent variable*?

- A) Predictor
- B) Feature
- C) **Target**
- D) Parameter

The goal of regression is to:

- A) Group similar data points
 - B) **Predict continuous outcomes**
 - C) Classify labels into categories
 - D) Reduce dimensionality
- Predict continuous outcomes

T / F

In a linear regression model, b_0 represents the slope of the line. **False** – the intercept.

Essay

Why do we minimize the loss function during training?

→ To find parameters that yield predictions closest to the true outputs.

THANK
you